UNIT 03: Digital and Analog Transmission

CONTENTS

Objectives

- represent digital data by using digital signals.
- learning schemes to transmit data digitally.
- learn the digital to analog conversions.
- understand the various modulation techniques.

Digital Transmission

A computer network is designed to send information from one point to another. This information needs to be converted to either a digital signal or an analog signal for transmission. In this chapter, we discuss the first choice, conversion to digital signals. First, we discuss digital-to-digital conversion techniques, methods which convert digital data to digital signals. Second, we discuss analog-to-digital conversion techniques, methods which change an analog signal to a digital signaL Finally, we discuss transmission modes

3.1 Digital-to Digital Transmission

We said that data can be either digital or analog. We also said that signals that represent data can also be digital or analog. In this section, we see how we can represent digital data by using digital signals. The conversion involves three techniques:

- Line coding,
- Block coding,
- Scrambling. Line coding is always needed~ block coding and scrambling mayor may not be needed.

Line Coding

Line coding is the process of converting digital data to digital signals. We assume that data, in the form of text, numbers, graphical images, audio, or video, are stored in computer memory as sequences of bits . Line coding converts a sequence of bits to a digital signal. At the sender, digital data are encoded into a digital signal; at the receiver, the digital data are recreated by decoding the digital signal. Figure 1 shows the process.

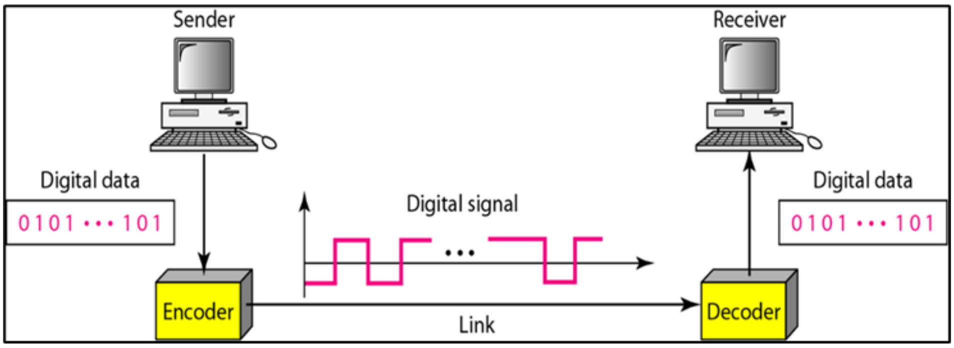


Figure 1 Digital to Digital Transmission

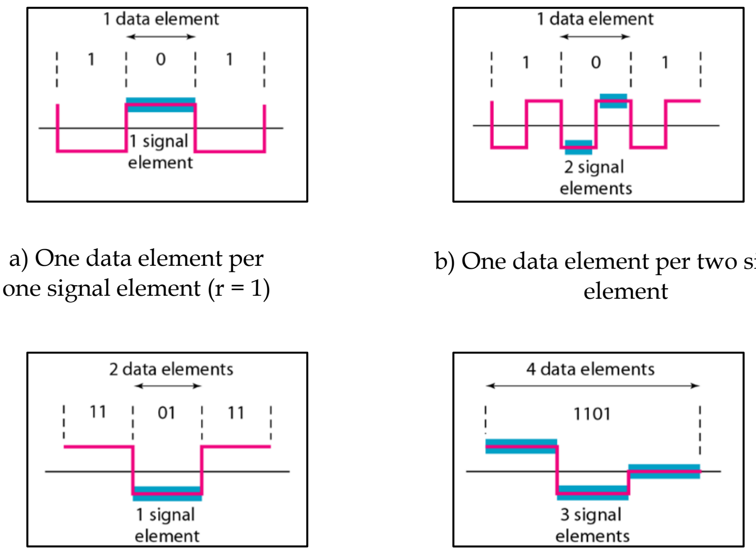
Characteristics

Before discussing different line coding schemes, we address their common characteristics.

Signal Element Versus Data Element

Let us distinguish between a data element and a signal element. In data communications, our goal is to send data elements. A data element is the smallest entity that can represent a piece of information: this is the bit. In digital data communications, a signal element carries data elements. A signal element is the shortest unit (timewise) of a digital signal. In other words, data elements are what we need to send; signal elements are what we can send. Data elements are being carried; signal elements are the carriers.

We define a ratio *r* which is the number of data elements carried by each signal element. Figure 2 shows several situations with different values of *r*.



a) One data element per one signal element (r = 1)

b) One data element per two signals element

c) Two data elements per one signal element (r = 2)

d) Four data elements per three signal elements (r = 4/3)

*Figure SEQ Figure * ARABIC 2 Signal element vs Data element*

In part a of the figure, one data element is carried by one signal element *(r* = 1). In part b of the figure, we need two signal elements (two transitions) to carry each data element *(r* = 1). We will see later that the extra signal element is needed to guarantee synchronization. In part c of the figure, a signal element carries two data elements *(r* = 2). Finally, in part d, a group of 4 bits is being carried by a group of three signal elements *(r* = 4/3). For every line coding scheme we discuss, we will give the value of *r*.

An analogy may help here. Suppose each data element is a person who needs to be carried from one place to another. We can think of a signal element as a vehicle that can carry people. When *r* =1, it means each person is driving a vehicle. When $r > 1$, it means more than one person is travelling in a vehicle (a carpool, for example). We can also have the case where one person is driving a car and a trailer ($r = 1/2$).

Data Rate V/s Signal Rate

The data rate defines the number of data elements (bits) sent in Is. The unit is bits per second (bps). The signal rate is the number of signal elements sent in Is. The unit is the baud. There are several common terminologies used in the literature. The data rate is sometimes called the bit rate; the signal rate is sometimes called the pulse rate, the modulation rate, or the baud rate.

One goal in data communications is to increase the data rate while decreasing the signal rate. Increasing the data rate increases the speed of transmission; decreasing the signal rate decreases the bandwidth requirement. In our vehicle-people analogy, we need to carry more people in fewer vehicles to prevent traffic jams. We have a limited *bandwidth* in our transportation system. The difference between datarate and signal rate is shown in Table 1.

Table 1 Data rate V/s Signal rate

Data rate	Signal rate
The number of data elements (bits) sent in 1s.	The number of signal elements sent in 1s.
The unit is bits per second (bps).	The unit is the baud.
Also known as bit rate.	Also known as pulse rate.

Baseline Wandering

In decoding a digital signal, the receiver calculates a running average of the received signal power. This average is called the *baseline.* The incoming signal power is evaluated against this baseline to determine the value of the data element. A long string of Os or 1s can cause a drift in the baseline (baseline wandering) and make it difficult for the receiver to decode correctly. A good line coding scheme needs to prevent baseline wandering.

DC Components

When the voltage level in a digital signal is constant for a while, the spectrum creates very low frequencies (results of Fourier analysis). These frequencies around zero, called DC (direct-current) *components,* present problems for a system that cannot pass low frequencies or a system that uses electrical coupling (via a transformer). For example, a telephone line cannot pass frequencies below 200 Hz. Also a long-distance link may use one or more transformers to isolate different parts of the line electrically. For these systems, we need a scheme with no DC component.

Self-synchronization

To correctly interpret the signals received from the sender, the receiver's bit intervals must correspond exactly to the sender's bit intervals. If the receiver clock is faster or slower, the bit intervals are not matched and the receiver might misinterpret the signals. A self-synchronizing digital signal includes timing information in the data being transmitted. This can be achieved if there are transitions in the signal that alert the receiver to the beginning, middle, or end of the pulse. If the receiver's clock is out of synchronization, these points can reset the clock.

Built-in Error Detection

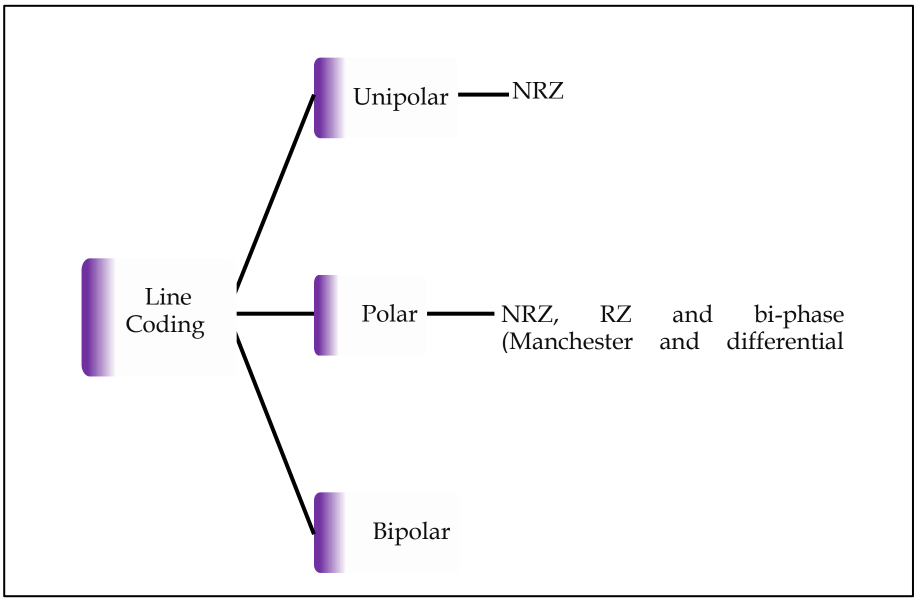
It is desirable to have a built-in error-detecting capability in the generated code to detect some of or all the errors that occurred during transmission. Some encoding schemes that we will discuss have this capability to some extent. Immunity to Noise and Interference Another desirable code characteristic is a code that is immune to noise and other interferences. Some encoding schemes that we will discuss have this capability.

Complexity

A complex scheme is more costly to implement than a simple one. For example, a scheme that uses four signal levels is more difficult to interpret than one that uses only two levels.

3.2 Line Coding Schemes

Line Coding schemes are divided into three categories as shown in Figure 3.



Unipolar

In a unipolar scheme, all the signal levels are on one side of the time axis, either above or below.

NRZ (Non-Return-to-Zero)

Traditionally, a unipolar scheme was designed as a non-return-to-zero (NRZ) scheme in which the positive voltage defines bit I and the zero voltage defines bit O. It is called NRZ because the signal does not return to zero at the middle of the bit. Figure 4 shows a unipolar NRZ scheme.

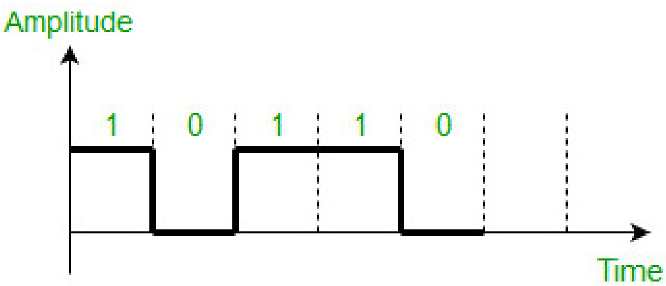


Figure 4 Unipolar NRZ Scheme

NRZ-L and NRZ-I – These are somewhat similar to unipolar NRZ scheme but here we use two levels of amplitude (voltages). For NRZ-L(NRZ-Level), the level of the voltage determines the value of the bit, typically binary 1 maps to logic-level high, and binary 0 maps to logic-level low, and for NRZ-I(NRZ-Invert), two-level signal has a transition at a boundary if the next bit that we are going to transmit is a logical 1, and does not have a transition if the next bit that we are going to transmit is a logical 0.

Note – For NRZ-I we are assuming in the example that previous signal before starting of data set "01001110" was positive. Therefore, there is no transition at the beginning and first bit "0" in current data set "01001110" is starting from +V. Example: Data = 01001110 is shown in Figure 5.

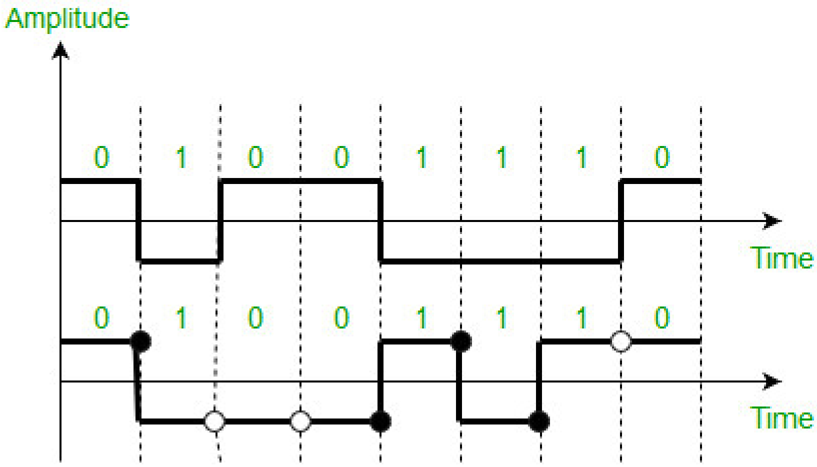


Figure 5 NRZ-L and NRZ-I

Comparison between NRZ-L and NRZ-I: Baseline wandering is a problem for both of them, but for NRZ-L it is twice as bad as compared to NRZ-I. This is because of transition at the boundary for NRZ-I (if the next bit that we are going to transmit is a logical 1). Similarly self-synchronization problem is similar in both for long sequence of 0's, but for long sequence of 1's it is more severe in NRZ-L.

Return to zero (RZ)

One solution to NRZ problem is the RZ scheme, which uses three values positive, negative, and zero. In this scheme signal goes to 0 in the middle of each bit.
Note – The logic we are using here to represent data is that for bit 1 half of the signal is represented by +V and half by zero voltage and for bit 0 half of the signal is represented by -V and half by zero voltage. Example: Data = 01001 as shown in Figure 6.

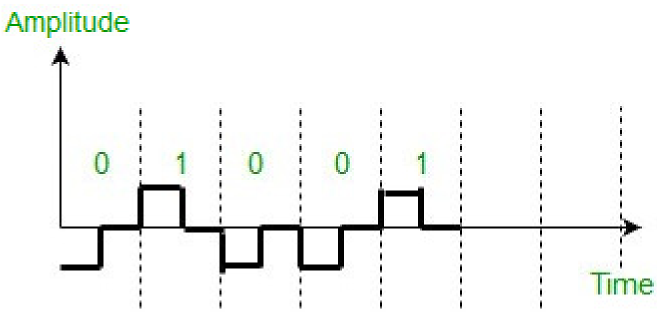


Figure 6 Return to Zero

Main disadvantage of RZ encoding is that it requires greater bandwidth. Another problem is the complexity as it uses three levels of voltage. As a result of all these deficiencies, this scheme is not used today. Instead, it has been replaced by the better-performing Manchester and differential Manchester schemes.

Biphase (Manchester and Differential Manchester)

- **Manchester encoding** is somewhat combination of the RZ (transition at the middle of the bit) and NRZ-L schemes. The duration of the bit is divided into two halves. The voltage remains at one level during the first half and moves to the other level in the second half. The transition at the middle of the bit provides synchronization.
- **Differential Manchester** is somewhat combination of the RZ and NRZ-I schemes. There is always a transition at the middle of the bit but the bit values are determined at the beginning of the bit. If the next bit is 0, there is a transition, if the next bit is 1, there is no transition.

The logic we are using here to represent data using Manchester is that for bit 1 there is transition form -V to +V volts in the middle of the bit and for bit 0 there is transition from +V to -V volts in the middle of the bit. For differential Manchester we are assuming in the example that previous signal before starting of data set "010011" was positive. Therefore there is transition at the beginning and first bit "0" in current data set "010011" is starting from -V. Example: Data = 010011.

The Manchester scheme overcomes several problems associated with NRZ-L, and differential Manchester overcomes several problems associated with NRZ-I as there is no baseline wandering and no DC component because each bit has a positive and negative voltage contribution.

Only limitation is that the minimum bandwidth of Manchester and differential Manchester is twice that of NRZ.

Bipolar schemes

In this scheme there are three voltage levels positive, negative, and zero as shown in *Figure 7*. The voltage level for one data element is at zero, while the voltage level for the other element alternates between positive and negative.

- ***Alternate Mark Inversion (AMI)***

  A neutral zero voltage represents binary 0. Binary 1's are represented by alternating positive and negative voltages.

- ***Pseudoternary***

  Bit 1 is encoded as a zero voltage and the bit 0 is encoded as alternating positive and negative voltages i.e., opposite of AMI scheme. Example: Data = 010010.

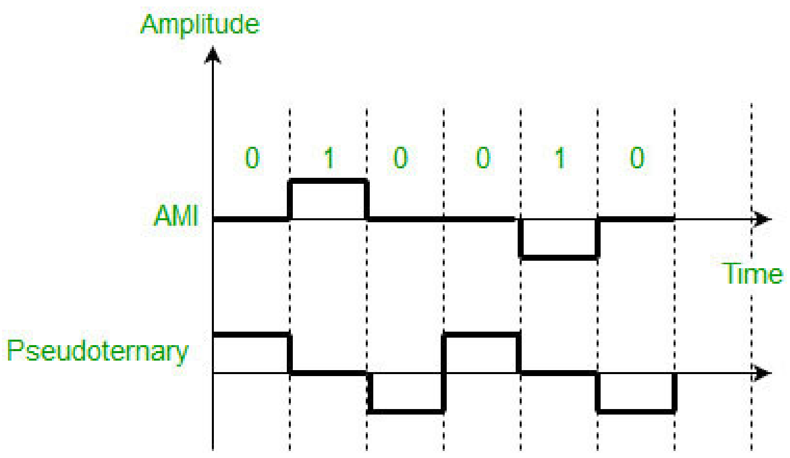


Figure 7: AMI and Pseudoternary

The bipolar scheme is an alternative to NRZ. This scheme has the same signal rate as NRZ, but there is no DC component as one bit is represented by voltage zero and other alternates every time .

Data Communication System

In data communication system, digital and analog communication together plays a very important and integrated role irrespective of many advantages of digital communications over analog. *Figure 8* and *Figure 9* shows the integrated role of digital and analog communication to complete data communication system. *Figure 9* shows that the link between modems is modulated analog signal created by the modem. The communication from PC to modem is consisted of binary signal

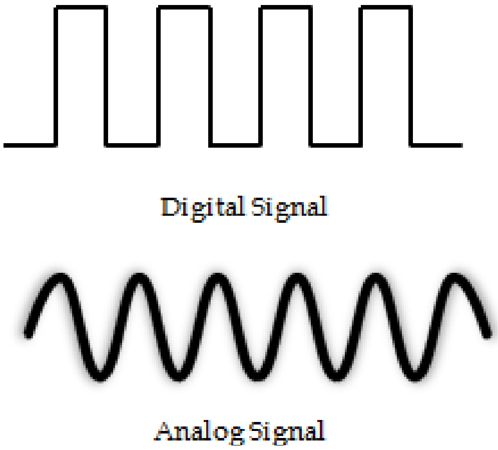


Figure SEQ Figure * ARABIC 8: Digital and Analog Signal

where as the communication between Central Telephone Office (CTO) and modem takes place in modulated analog signal. The communication between CTO to another CTO is by digital signal using time division multiplexers. Thereafter CTO feeds modulated analog signal to modem and modem converts it into binary signal for the PC. We may now say that different types of signals emerging on the communication link and reaching to CTO on their way across a city. These can be multiplexed to share the same communication link for transmitting to destination.

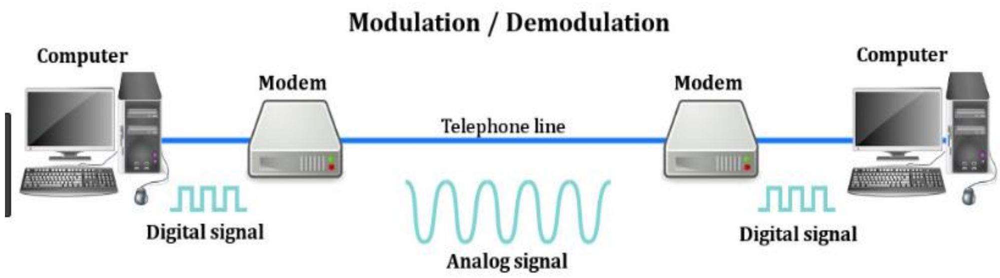


Figure 9: Data Communication System

3.3 Circuits, Channels and Multichanneling

A circuit is a path between two or more points along which an electrical current flows. In data communication, a circuit is considered as a specific path between two or more points along which signals is carried. The signals may be analog, binary or digital. This has been shown in the Figures 9 and 10.

3.4 Modulation of Digital Signal

A digital transmission uses low pass channel with high bandwidth. Similarly, analog transmission is also possible on band pass channels that require converting of binary data or a low pass analog signal into a band pass analog signal. This technique is called modulation. Thus, modulation of binary data or digital to analog modulation is carried over by changing one of the characteristics of the analog signal in accordance with the information in the digital signal. The information in the digital signal is always in the form of 0 and 1. The characteristics of analog signal that are altered are amplitude, frequency and phase of the analog waveform. Based on the change in one of the characteristics, the digital to analog modulation may be of amplitude shift keying (ASK), frequency shift keying (FSK) and phase shift keying (PSK) types. Quadrature amplitude modulation is the fourth category that combines changes in both amplitude and phase to provide better efficiency.

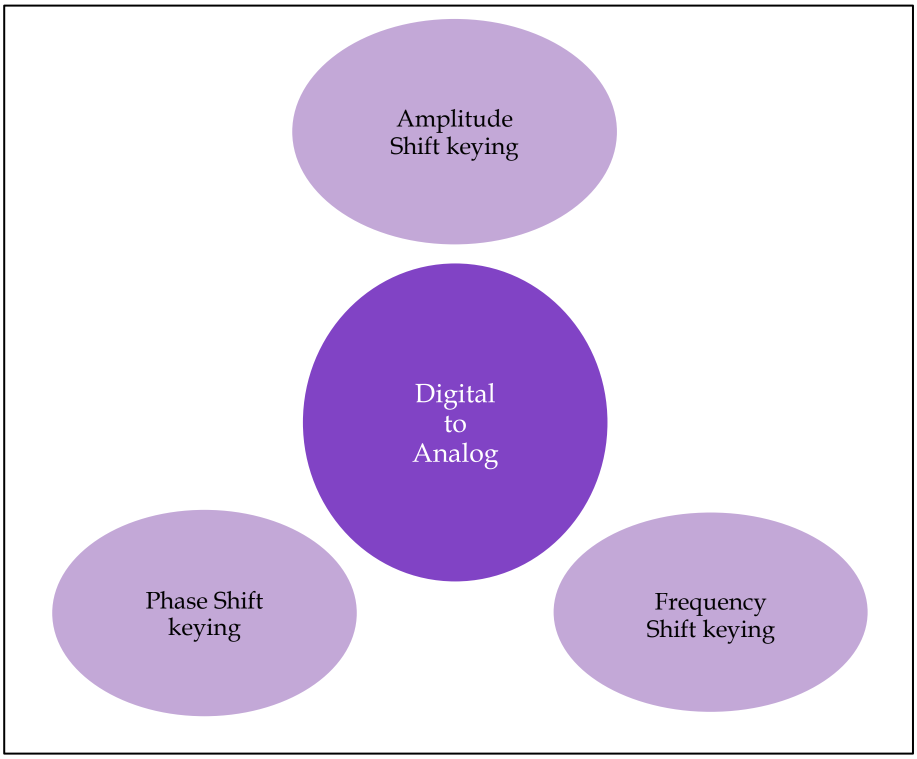


Figure 10: Techniques used for Digital to Analog Conversion

Data Rate

Bit rate is the number of bits (0 or 1) transmitted during 1 second of time. The number of signal changes per unit of time to represent the bits is called the data rate of the modem. That rate is usually expressed in terms of a unit known as a baud. A signal unit may have 1 or more than 1 bits. Therefore, the baud is the number of times per second the line condition can switch from "1" to "0". Baud rate and bit rate, which are expressed in bits per second, usually are not the same, as several bits may be transmitted through the channel by the modem in each signal change (a few bits can be transmitted as one symbol). The relation between bit rate and baud is expressed that bit rate equals the baud rate times the number of bits represented by each signal unit. Bit rate is always more or equal than baud rate. The reason for baud rate is that it determines the bandwidth required to transmit the signal. The signal may be in the form of pieces or block that may contain bits. A fewer bandwidth required to move these signal unit with large bits for an efficient system. To understand the relation between bit and baud rate, we consider an analogy of car, passengers and highway with signal units, bits and bandwidth respectively. A car has capacity of carrying 5 passengers maximum at a time. Suppose a highway may support only 1000 cars per unit time without congestion. When each car on the highway carries 5 passengers, it is considered that the highway is capable of providing services without congestion. Thus highways services are treated efficient. Consider another case, when all these 5000 passengers wish to go in separate cars, they require 5000 cars and highway can only support 1000 cars at a time. The services offered get deteriorated because highway's capacity is meant only for 1000 cars. It does not bother as to whether these 1000 cars are carrying 1000 passengers or 5000 passengers or more. To support more cars, the highway needs to widen. Similarly, the number of bauds determines the bandwidth.

Carrier Signal

The carrier signal that is a high frequency signal plays a significant role in the modulation and data transmission. It is the base signal generated by the sending device whose one of the characteristics is altered in accordance with the digital signal to be modulated. The modulating signal or digital signal riding over the carrier signal is transmitted to the receiving device. The receiving device is tuned to the frequency of the carrier signal. Other advantages of the carrier signal are that it provides efficient transmission between sending device and receiving device and needs smaller sizes of antenna because of higher frequency of transmission.

Amplitude Shift Keying (ASK)

ASK describes the technique how the carrier wave is multiplied by the digital signal f (t) so that the strength of the carrier wave is varied to represent binary 0 and 1. In ASK, both the frequency and phase of an analog waveform are kept uniform while amplitude is changed in accordance with the digital signal.

Mathematically, the modulated carrier signal y(t) is:

y(t) = f(t) × sin(2pfct + j) where fc is a carrier frequency and t is instantaneous time. The following Figure 11 represents ASK modulated waveform along with its input

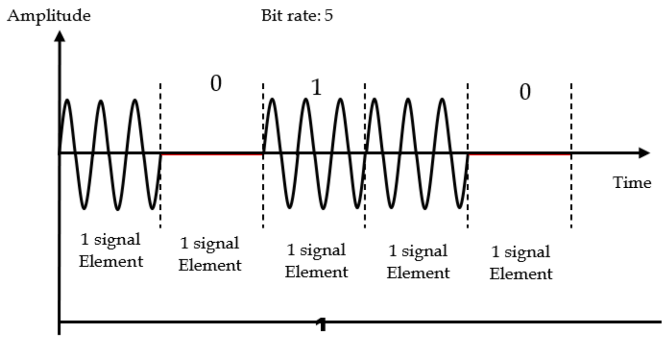


Figure 11 Binary ASK or ON-OFF Keying

The main advantage of ASK is that it is easy to produce and detect. The disadvantages of ASK are that it is highly susceptible to noise interference that changes the amplitude of the signal. A 0 can be changed to 1 and vice versa. Other drawbacks are that the speed of the changing amplitude is limited by the bandwidth of the line and the small amplitude changes suffer from unreliable detection. Telephone lines limit amplitude changes to some 3000 changes per second. The disadvantages of amplitude modulation causes this technique to no longer be used by modems, however, it is used in conjunction with other techniques.

The bandwidth for an ASK signal is mathematically given by:

Bandwidth (BW) = (1 + d) × Nbaud

Where Nbaud is the baud rate and d is the modulating index and may have minimum value as 0.

Frequency Shift Keying

Frequency of analog carrier signal is modified in accordance with the message signal. FSK describes the modulation of a carrier (or two carriers) by using a different frequency for a 1 or 0. In this technique the frequency of the carrier signal is changed according to the data while keeping the amplitude and phase constant. The transmitter sends different frequencies for a 1 than for a 0 as shown in *Figure 12*. The resultant modulated signal may be regarded as the sum of two amplitude modulated signals of different carrier frequency.

Mathematically, the modulated wave y(t) can be shown as y(t) = f1(t) sin(2pfc1t + j) + f2(t) sin(2pfc2t + j) where fc1 and fc2 are different carrier frequencies of two different signals. FSK is classified as wide band if the separation between the two carrier frequencies is larger than the bandwidth of the spectrums. Narrow-band FSK is the term used to describe an FSK signal whose carrier frequencies are separated by less than the width of the spectrum than ASK for the same modulation.

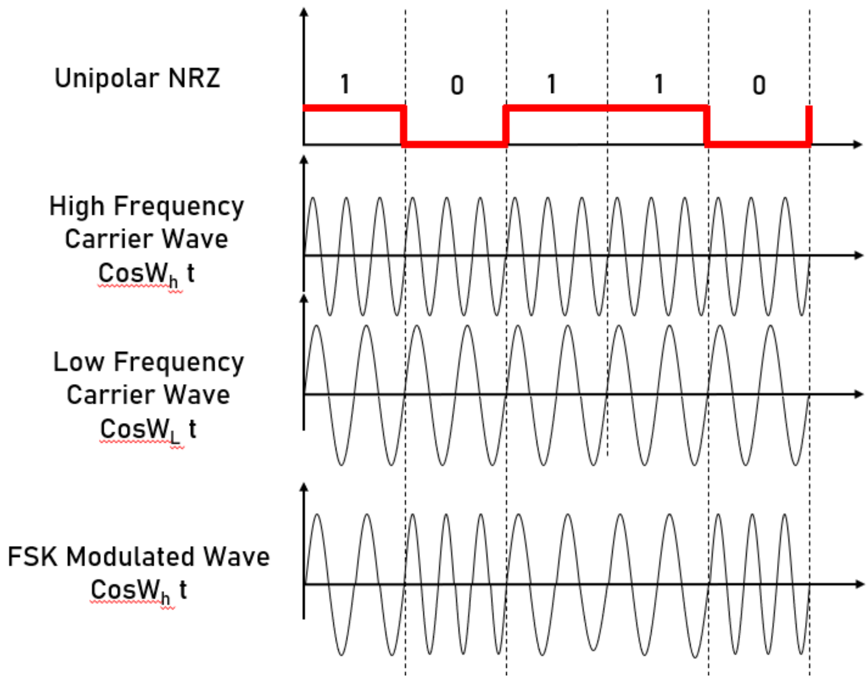


Figure 12 Frequency Shift Keying

The advantage of FSK is that it provides better immunity from noise because the receiving device looks for specific frequency changes over given number of periods and frequency is almost unaffected from noise. The disadvantages of this technique are that again as it was with amplitude modulation. The rate of frequency changes is limited by the bandwidth of the line, and that distortion caused by the lines makes the detection even harder than amplitude modulation. Today this technique is used in low rate asynchronous modems up to 1200 baud only. The bandwidth for FSK signal is the sum of the baud rate of the signal and the frequency shift. The frequency shift is the difference between the two carrier frequencies.

Phase Shift Keying

In this modulation method a sine wave is transmitted and the phase of the sine wave carries the digital data or the phase of sine wave is varied to represent binary 1 or 0 and both the amplitude and frequency of the analog waveform are kept constant. For a 0, a 0 degrees phase sine wave is transmitted. For a 1, a 180 degrees sine wave is transmitted. As this method involves two states of phase changes, it is called binary PSK or 2-PSK. This technique, in order to detect the phase of each symbol, requires phase synchronization between the receiver's and transmitter's phase. This complicates the receiver's design. The advantages of PSK are that it is immune to noise and is not band limited.

Differential Phase Modulation A sub method of the phase modulation is differential phase modulation. In this method, the modem shifts the phase of each succeeding signal in a certain number of degrees for example, a 0 for 90 degrees and 1 for 270 degrees as illustrated in *Figure 13*.

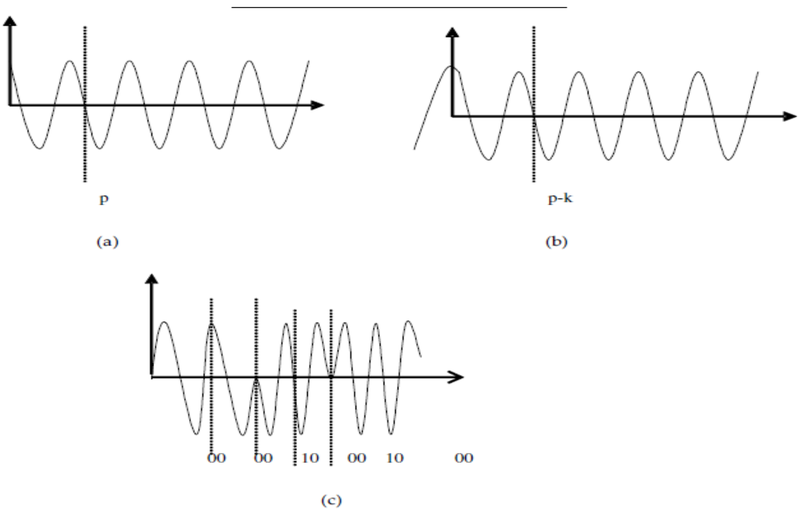


Figure 13: Phase Shift Keying

PSK is a technique, which shifts the period of a wave. The wave shown in Figure 13 (a) has a period of p starting from 0. The wave shown in Figure 13 (b) is the same wave as shown in Figure 13 (c), but its phase has been shifted. Notice that the period starts at the wave's highest point 1 on the vertical axis. It just so happens that we have shifted this wave by one quarter of the wave's full period. We can shift it another quarter, if we want to, so the original wave would be shifted by half of its period. And we could do it one more time, so that it would be shifted three quarters of its original period. This means there exist 4 separate waves and therefore each wave is provided for some binary value. Since there are 4, 2 bits are provided to each wave which is represented below Table 2.

Table 2 Bit Value and Amount of Shift

Bit Value	Amount of Shift
00	None
01	¼
10	½
11	¾

This technique of letting each shift of a wave represent some bit value is phase shift keying. But the real key is to shift each wave relative to the wave that came before it. PSK describes the modulation technique that alters the phase of the carrier. Mathematically, it can be represented as $y(t) = f(t) \sin(2pfct + j(t))$ where jc is phase shift. This method is easier to detect than the previous one. The receiver has to detect the phase shifts between symbols and not the absolute phase.

Binary Phase Shift Keying (BPSK): In the case of two possible phases shift the modulation will be called BPSK - binary PSK. In the case of 4 different phase shifts possibilities for each symbol which means that each symbol represents 2 bits the modulation will be called quadrature PSK (QPSK), and in case of 8 different phase shifts the modulation technique will be called 8-PSK. A single data channel modulates the carrier. A single bit transition, 1 to 0 or 0 to 1, causes a 180-degree phase shift in the carrier. Thus, the carrier is said to be modulated by the data. As this has only two phases, 0 and 1 as shown in *Figure 14* and *Figure 15*. It is therefore a type of ASK with taking the values -1 or 1 and its bandwidth is the same as that of ASK. Phase shift keying offers a simple way of increasing the number of levels in the transmission without increasing the bandwidth by introducing smaller phase shifts. Quadrature phase-shift-keying (QPSK) has four phases such as 0, p/2, p, 3p/2. Consequently, M-ary PSK has M phases given by 2pm/M; m = 0,1…M-1. For a given bit-rate, QPSK requires half the bandwidth of PSK and is widely used for this reason.

The number of times the signal parameter (amplitude, frequency, and phase) is changed per second is called the signaling rate. It is measured in baud. 1 baud = 1 change per second. With binary modulations such as ASK, FSK and BPSK, the signaling rate equals the bit-rate. With QPSK and M-ary PSK, the bit-rate may exceed the baud rate.

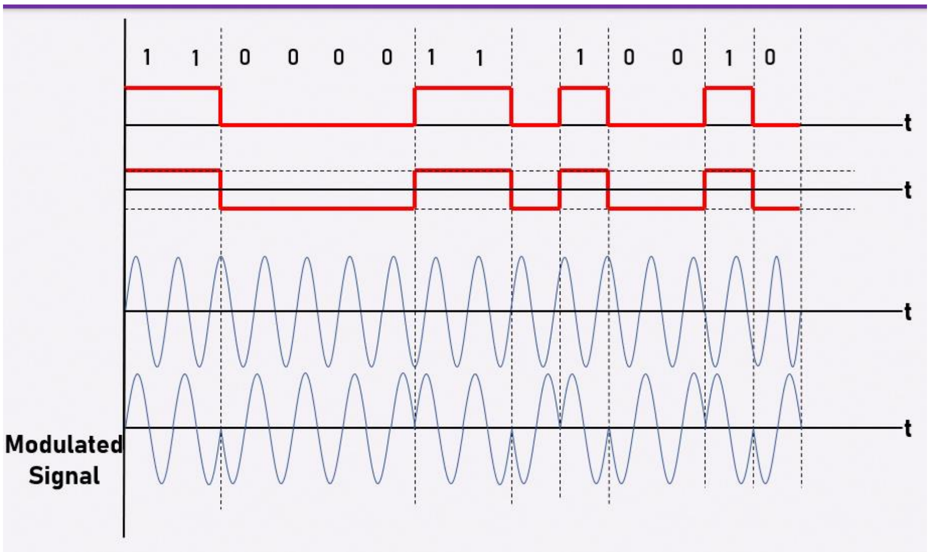


Figure 14 Binary Phase Shift Keying

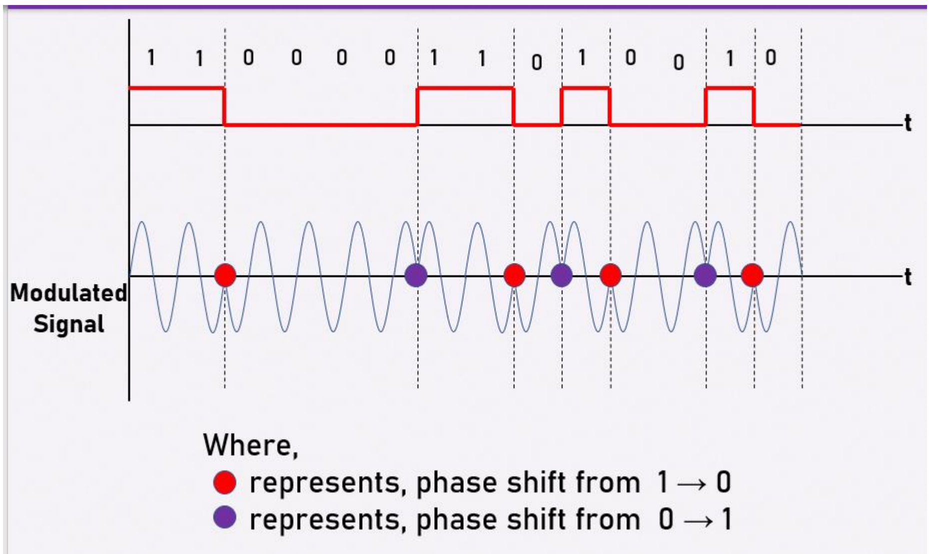


Figure 15: Output Modulated Signal

Quadrature Phase Shift Keying (QPSK)

Two data channels modulate the carrier. Transitions in the data cause the carrier to shift by either 90 or 180 degrees. This allows transmission of two discrete data streams, identified as I channel (In phase) and Q channel (Quadrature) data. In this method four different phase angles are used.

In QPSK, the four angles are usually out of phase by 90°.

QPSK Modulator

Quadrature Phase Shift Keying (**QPSK**) is a form of Phase Shift Keying in which two bits are **modulated** at once, selecting one of four possible carrier phase shifts (0, 90, 180, or 270 degrees). **QPSK** allows the signal to carry twice as much information as ordinary PSK using the same bandwidth

Analog to Analog Conversion

Analog-to-analog conversion, or analog modulation, is the representation of analog information by an analog signal. An example is radio. Analog to analog can be represented in three ways: Amplitude Modulation, Frequency Modulation and Phase Modulation.

3.5 Amplitude Modulation

This describes the technique in which the carrier wave is multiplied by the digital signal f(t). Mathematically, the modulated carrier signal y(t) is: y(t) = f(t) sin(2p¦ct+j) where fc is a carrier frequency and t is instantaneous time. *Figure 16* shows the technique of amplitude modulation. The main advantage of this technique is that it is easy to produce such signals and also to detect them. This technique has two major disadvantages. The first is that the speed of the changing amplitude is limited by the bandwidth of the line. The second is that the small amplitude changes suffer from unreliable detection. Telephone lines limit amplitude changes to some 3000 changes per second. The disadvantages of amplitude modulation causes this technique to no longer be used by modems, however, it is used in conjunction with other techniques.

QAM (Quadrature Amplitude Modulation)

This technique is based on the basic amplitude modulation. This technique improves the performance of the basic amplitude modulation. In this technique two carrier signals are transmitted simultaneously. The two carrier signals are at the same frequency with a 90 degrees phase shift.

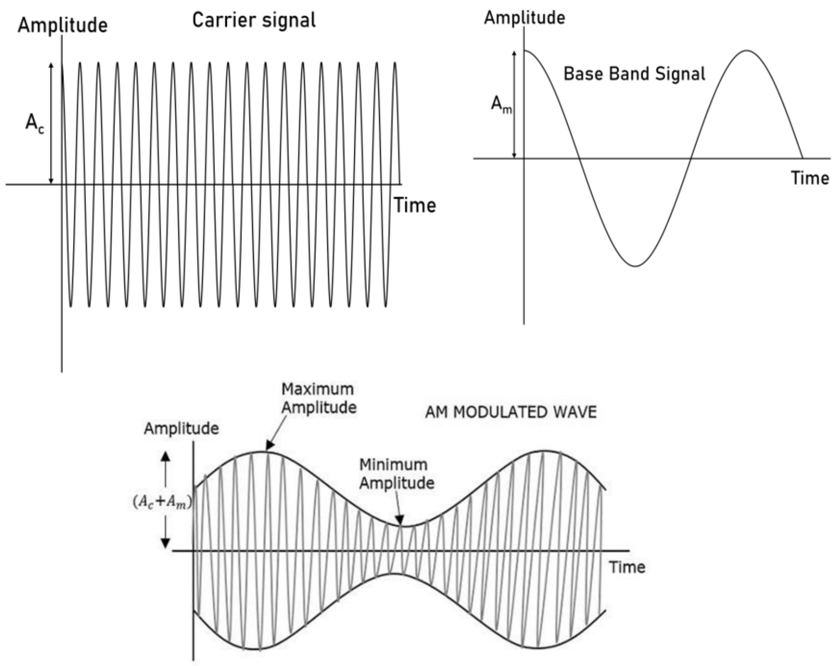


Figure 16: Amplitude Modulation

Advantages and Disadvantages of Amplitude modulation

In order that a radio signal can carry audio or other information for broadcasting or for two-way radio communication, it must be modulated or changed in some way.

Advantages and disadvantages of amplitude modulation are shown below in *Table 3*.

Table 3 Advantages and Disadvantages of Amplitude Modulation

Advantages	Disadvantages
It is simple to implement	It is not efficient in terms of its power usage
It can be demodulated using a circuit consisting of very few components	It is not efficient in terms of its use of bandwidth, requiring a bandwidth equal to twice that of the highest audio frequency.

AM receivers are very cheap as no specialized components are needed.	It is prone to high levels of noise because most noise is amplitude based and obviously AM detectors are sensitive to it.

3.6 Frequency Modulation

Frequency Modulation involves the modulation of the frequency of the analog sine wave where the instantaneous frequency of the carrier is deviated in proportion of the deviation of the modulated carrier with respect to the frequency of the instantaneous amplitude of the modulating signal. It may be said in a simple word that it occurs when the frequency of a carrier is changed based upon the amplitude of input signal. Unlike AM, the amplitude of carrier signal is unchanged. This makes FM modulation more immune to noise than AM and improves the overall signal-to noise ratio of the communications system. Power output is also constant, differing from the varying AM power output. The amount of analog bandwidth necessary to transmit FM signal is greater than the mount necessary for AM, a limiting constraint for some systems. The modulating index for FM is given as below :

$$\beta = fp/fm,$$

where

β *= Modulation index,* f_m *= frequency of the modulating signal and* f_p *= peak frequency deviation*.

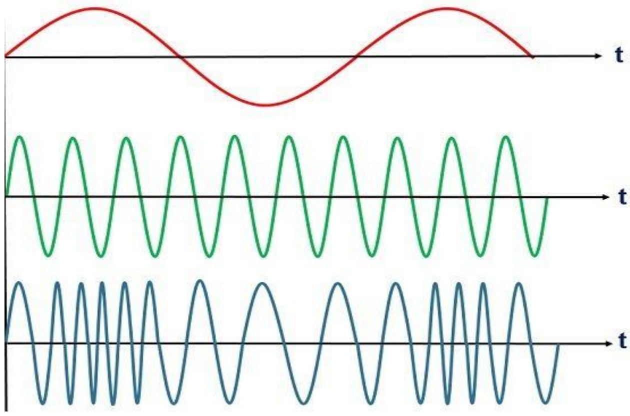


Figure 17: Frequency Modulation

From the *Figure* 17, it is inferred that the amplitude of the modulated signal always remains constant, irrespective of frequency and amplitude of modulating signal. It means that the modulating signal adds no power to the carrier in frequency modulation unlike to amplitude modulation. FM produces an infinite number of side bands spaced by the modulation frequency, fm that is not in case of AM. Therefore, AM considered a linear process whereas FM as a nonlinear process. It is necessary to transmit all side bands to reproduce a distortion free signal. Ideally, the bandwidth of the modulated signal is infinite in this case. In general the determination of the frequency content of an FM waveform is complicated, but when b is small, the bandwidth of the FM signal is 2fm. On the other hand, when b is large, the bandwidth is determined (empirically) as 2 fm (1 + b).

The difference between amplitude modulation and frequency modulation are shown in Table 4.

Table 4. Difference between Amplitude Modulation and Frequency Modulation

AMPLITUDE MODULATION	FREQUENCY MODULATION
In amplitude modulation, the frequency and phase remain the same.	In frequency modulation amplitude and phase remain the same.

Its modulation index varies from 0 to 1.	Its modulation index is always greater than one.
It has only two sidebands.	It has an infinite number of sidebands.
It has simple circuit.	It has complex circuit.
The amplitude of the carrier wave is modified in order to send the data or information.	The frequency of the carrier wave is modified in order to send the data or information.
The amplitude of the carrier wave is modified in order to send the data or information.	The frequency of the carrier wave is modified in order to send the data or information.
It requires low bandwidth in the range of 10 kHz.	It requires high bandwidth in the range of 200 kHz.
It works in a frequency range of 535 to 1705 Kilohertz (KHz).	It works in a frequency range of 88 to 108 Megahertz (MHz).
It operates in the medium frequency (MF) and high frequency (HF).	It operates in the very high frequency.
It has poor sound quality.	It has better sound quality.

Phase Modulation

Phase Modulation (PM) is similar to frequency modulation. Instead of the frequency of the carrier wave changing, the phase of the carrier wave changes. In PM the phase of the carrier is made proportional to the instantaneous amplitude of the modulating signal. Modulating index for PM is given as b = Dj, where Dj is the peak phase deviation in radians. As in the case of angular modulation argument of sinusoidal is varied and therefore we will have the same resultant signal properties for frequency and phase modulation. A distinction in this case can be made only by direct comparison of the signal with the modulating signal wave, as shown in *Figure 18*.

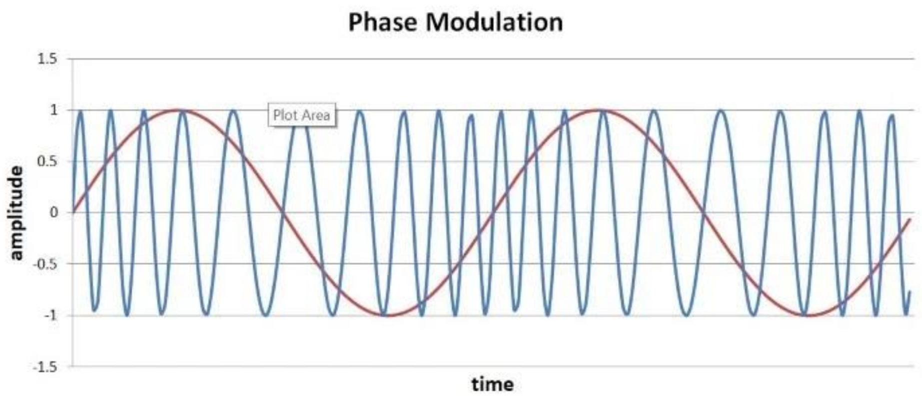


Figure 18: Phase Modulation

Caution Phase modulation and frequency modulation are interchangeable by selecting the frequency response of the modulator so that its output voltage will be proportional to integration of the modulating signal and differentiation of the modulating signal respectively. Bandwidth and power issues are same as that of the frequency modulation.

3.8 Self Assessment

State whether the following statements are true or false:

1. The modulation is the act of translating some high-frequency (base band signal) such as voice, data, etc. to a lower frequency.

2. Amplitude Modulation (AM) involves the modulation of the amplitude of the carrier as analog sine wave.

3. FM considered a linear process whereas AM as a nonlinear process.

4. In PM the phase of the carrier is made proportional to the instantaneous amplitude of the modulating signal.

5. Analog to analog signal conversion involves amplitude modulation and frequency modulation techniques only.

6. The process of changing one of the characteristics of a carrier analog signal based on the information in a digital signal is called ___________ conversion

a. analog-to-analog
b. analog-to-digital
c. digital-to-analog
d. digital-to-digital

7. In ______________ the frequency of the carrier signal is varied based on the information in a digital signal.

a. ASK
b. PSK
c. FSK
d. QAM

8. In _____________ the amplitude of the carrier signal is varied based on the information in a digital signal.

a. ASK
b. PSK
c. FSK
d. QAM

9. In ___________ the phase of the carrier signal is varied based on the information in a digital signal.

a. ASK
b. PSK
c. FSK
d. QAM

10. Most modern modems use ___________ for digital to analog modulation.

a. ASK
b. PSK
c. FSK
d. QAM

11. ________ rate is the number of bits per second; ____________ rate is the number of signals unit per second.

a. Baud; bit

b. Bit; baud
c. Baud; base
d. Base; baud

12. For ___________, the minimum bandwidth required for transmission is equal to the baud rate.

a. ASK
b. PSK
c. FSK
d. a and b

13. In which type of modulation can the bit rate be four times the baud rate ?

a. ASK
b. FSK
c. PSK
d. None of the above

14. In which type of modulation can the bit rate be three times the baud rate?

a. ASK
b. FSK
c. PSK
d. none of the above

15. In which type of modulation can the bit rate be half the baud rate.

a. ASK
b. FSK
c. PSK
d. None of the above.

16. ASK,PSK, FSK and QAM are examples of ______________ conversion.

a. digital-to-digital
b. digital-to-analog
c. analog-to-analog
d. analog-to-digital

17. ___________ conversion is the process of changing one of the characteristics of an analog signal based on the information in the digital data.

a. Digital–to-analog
b. Analog-to-analog
c. Analog-to-digital
d. Digital-to-digital

Answers:

1. False	2. True	3. False
4. True	5. False	6. digital-to-analog
7. FSK	8. PSK	9. PSK
10. QAM	11. Bit; baud	12. a and b
13. PSK	14. None of the above	15. None of the above
16. digital-to-analog	17. Digital-to-analog	

Summary

- A circuit is a path between two or more points along which signals is carried. The circuit may be a physical path consisting of wires or it may be wireless. A network, which is wired or wireless involves a number of circuits consisting of a number of intermediate switches.
- A virtual circuit is a logical path selected out of many possible physical paths available between two or more points.
- Multiplexing is the process in which multiple channels are combined for transmission over a common transmission path.
- The digital transmission requires a low pass channel with high bandwidth. The analog transmission can be carried on band pass channels. The different methods that convert binary data or a low pass analog signal into a band pass analog signal is called modulation.
- The digital to analog conversion includes ASK (Amplitude Shift Keying), FSK (Frequency Shift Keying), PSK (Phase Shift Keying), QPSK (Quadrature Phase Shift Keying), QAM (Quadrature Amplitude Modulation) and have been explained under the section Modem Modulation Techniques.
- Analog to analog signal conversion involves amplitude modulation, frequency modulation and phase modulation techniques.

Keywords

Amplitude Modulation: It involves the modulation of the amplitude of the carrier as analog sine wave.

Amplitude Shift Keying: ASK refers to technique how the carrier wave is multiplied by the digital signal f(t) so that the strength of the carrier wave is varied to represent binary 0 and 1.

Baud Rate: It is the number of times per second the line condition can switch from "1" to "0".

Binary Phase Shift Keying: BPSK involves two possible phases shift for the modulation.

Carrier Signal: It is the base signal generated by the sending device whose one of the characteristics is altered in accordance with the digital signal to be modulated.

Differential Phase Shift Keying: In this method, the modem shifts the phase of each succeeding signal in a certain number of degrees.

FDM: In frequency division multiplexing, multiple channels are combined together for transmission over a single channel.

FDMA: This divides the frequency band into various channels based on the FDM techniques. Each of these can carry a voice conversation or, with digital service, carry digital data.

Frequency Modulation: Frequency Modulation involves the modulation of the frequency of the analog sine wave.

Frequency Shifted Keying: FSK describes the modulation of a carrier (or two carriers) by using a different frequency for a 1 or 0.

Inter modulation: It involves two (or more) sinusoids effect one another to produce undesired products, that is, unwanted frequencies (noise).

Modems: Refers to the modulator and demodulator that converts analog signal to digital signal and vice versa.

Modulation: It is the act of translating some low-frequency (base band signal) such as voice, data, etc. to a higher frequency.

Multiplexing: Refers to the process in which multiple channels are combined for transmission over a common transmission path.

Phase Modulation: Phase Modulation (PM) is similar to frequency modulation. Instead of the frequency of the carrier wave changing, the phase of the carrier wave changes.

Phase Shift Keying: In this modulation method a sine wave is transmitted and the phase of the sine wave carries the digital data or the phase of sine wave is varied to represent binary 1 or 0 and both the amplitude and frequency of the analog waveform are kept constant.

Quadrature Amplitude Modulation: This technique is based on the amplitude modulation and phase modulation to improve the performance of the amplitude modulation.

Quadrature Phase Shifted Keying: In the case of 4 different phase shifts possibilities for each symbol which means that each symbol represents 2 bits the modulation will be called quadrature PSK (QPSK).

Space Division Switching: Refers to the kind of switch developed for analog environment. Crossbar switch is the simplest possible space division switch where each packet takes a different path through the switch depending on its destination. Time division switching is based on multiplexing for digital transmission.

TDM: Refers to the process to merge data from several sources into a single channel for communication over transmission media like telephone lines, microwave system or satellite system.

TDMA: This is a digital transmission technology that allows a number of channels to access a single radio frequency (RF) channel without interference by allocating unique time slots to each channel.

Virtual circuit: This is a logical path selected out of many possible physical paths available between two or more points.

WDM: This is defined as the fibre-optic transmission technique that employs two or more optical signals having different wavelengths to transmit data simultaneously in the same direction over one fibre, and later on is separated by wavelength at the distant end.

Further/Suggested Readings

- Andrew S. Tanenbaum, *Computer Networks*, Prentice Hall.
- Behrouz A. Forouzan and Sophia Chung Fegan, *Data Communications and Networking*, McGraw-Hill Companies.
- Burton, Bill, *Remote Access for Cisco Networks*, McGraw-Hill Osborne Media, McGraw-Hill Osborne Media.
- Rajneesh Agrawal and Bharat Bhushan Tiwari, *Computer Networks and Internet*, Vikas Publication

- https://www.geeksforgeeks.org/computer-network-tutorials/